SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: CSA51
COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Analytical Problem Solving: 40%

QUESTIONS: **Total Marks: 50**

Annexure A

Code of Conduct:

I Gurusurya.G(192311332)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Gurusurya

Signature of the student:

Annexure A
Analytical Problem Solving

1. RSA Encryption Analysis
Given p=11, q=13

a. Compute n ϕ(n)\
b. Choose e=7 and find the private key d
c. Encrypt the message M=9
d. Decrypt the ciphertext and verify the result
e. Analyze how changing e affects security

2. Block Cipher Mode Evaluation
Given plaintext blocks:
P=[1010,1010,1111,1010], Key = K=1100

a. Encrypt using ECB and CBC modes (Assume IV = 0000)
b. Compare the ciphertext outputs
c. Analyze which mode hides patterns better and explain why

3. DES vs 3-DES vs Blowfish Performance Study
Given:

a. DES key size = 56 bits, time = 2 ms/block
b. 3-DES key size = 168 bits, time = 6 ms/block
c. Blowfish key size = 128 bits, time = 1 ms/block

For 1000 blocks of data:

i. Calculate total encryption time for each algorithm
ii. Analyze trade-offs between security and performance
iii. Recommend the best algorithm for secure real-time communication

4. Diffie-Hellman Key Exchange Analysis
Given:

a. p=23, g=5
b. User A private key = 6
c. User B private key = 15
d. Compute public keys for A and B
e. Compute the shared secret key
f. Demonstrate how a Man-in-the-Middle attacker could alter the exchange
g. Suggest a secure method to prevent the attack

5. ECC vs RSA Comparison
Given:

a. RSA key size = 1024 bits
b. ECC key size = 160 bits (equivalent security)
c. Device processing capacity = limited (IoT device)
d. Analyze computational cost for both systems
e. Estimate which consumes less power and memory
f. Justify which cryptosystem is better for the given scenario with reasoning

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
		strong justification			
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

1. RSA Encryption Analysis
Given:

- $p = 11$
- $q = 13$
- $e = 7$
- Message $M = 9$

(a) Compute $\boldsymbol{n}$and $\boldsymbol{\phi}(\boldsymbol{n})$

$$n = p \times q = 11 \times 13 = 143$$

Euler's Totient:

$$\begin{aligned}&\phi(n) = (p-1)(q-1)\\&= (11-1)(13-1)\\&= 10 \times 12 = 120\end{aligned}$$

Answer:

- $n = 143$
- $\phi(n) = 120$

(b) Find the private key $\boldsymbol{d}$
We know

$$e \times d \equiv 1 \pmod{120}$$

Since

$$\begin{aligned}&7 \times 103 = 721\\&721 \div 120 = 6 \text{ remainder } 1\end{aligned}$$

Therefore,

$$d = 103$$

Keys

- Public Key = (7,143)
- Private Key = (103,143)

(c) Encrypt $\boldsymbol{M = 9}$
RSA Encryption:

$$\begin{aligned}&C = M^e n\\&= 9^7 143\end{aligned}$$

Using modular arithmetic,

$$\begin{aligned}&9^2 = 81\\&9^4 = 81^2 = 6561 \equiv 126\\&9^7 = 9^4 \times 9^2 \times 9\\&= 126 \times 81 \times 9\\&126 \times 81 = 10206 \equiv 53\\&53 \times 9 = 477 \equiv 48\end{aligned}$$

Ciphertext:

$$\boxed{C = 48}$$

(d) Decrypt the Ciphertext

$$\begin{aligned}&M = C^d n\\&= 48^{103} 143\end{aligned}$$

Using modular exponentiation,

$$48^{103} 143 = 9$$

Recovered plaintext:

$$\boxed{M = 9}$$

Verification:
Original Message = 9 ✓
Recovered Message = 9 ✓
Hence RSA works correctly.

(e) Effect of Changing e

Small e	**Large e**
Faster encryption	Slower encryption
Faster signature verification	Slower verification
Must satisfy gcd(e, φ)=1	Must satisfy gcd(e, φ)=1
Very small values (e=3) may introduce attacks if padding is poor	More computation but generally safe

Conclusion:

- Common choice is **65537** because it provides a good balance between speed and security.

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2. Block Cipher Mode Evaluation
Given
Plaintext Blocks

$$P = [1010, 1010, 1111, 1010]$$

Key

$$K = 1100$$

IV

$$0000$$

Assume encryption is simple XOR with the key.

(a) ECB Mode
Encryption:

$$C = P \oplus K$$

Block 1
1010
$\oplus$1100
=0110
Block 2
1010
$\oplus$1100
=0110
Block 3
1111
$\oplus$1100
=0011
Block 4
1010
$\oplus$1100
=0110
ECB Ciphertext

$$\boxed{[0110, \ ,0110, \ 0011 \ 0110]}$$

CBC Mode
Formula

$$C_i = (P_i \oplus C_{i-1}) \oplus K$$

Initial

$$C_0 = IV = 0000$$

Block 1
1010⊕0000
=1010
1010⊕1100
=0110
C1=0110

Block 2
1010⊕0110
=1100
1100⊕1100
=0000
C2=0000

Block 3
1111⊕0000
=1111
1111⊕1100
=0011
C3=0011

Block 4
1010⊕0011
=1001
1001⊕1100
=0101
C4=0101
CBC Ciphertext

$$\boxed{[0110, \ ,0000, \ 0011 \ 0101]}$$

(b) Comparison
ECB:
0110 0110 0011 0110
CBC:
0110 0000 0011 0101
Notice that repeated plaintext blocks in ECB produce identical ciphertext blocks, while CBC generates different ciphertext because each block depends on the previous ciphertext block.

(c) Which Hides Patterns Better?
ECB

- Same plaintext → Same ciphertext
- Reveals data patterns
- Not recommended for large files

CBC

- Uses previous ciphertext block
- Same plaintext produces different ciphertext
- Better confidentiality

Answer: CBC hides patterns better because chaining introduces randomness.

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3. DES vs 3-DES vs Blowfish Performance Study
Given
Algorithm Key Size Time/Block

Algorithm	**Key Size**	**Time/Block**
DES	56 bits	2 ms
3-DES	168 bits	6 ms
Blowfish	128 bits	1 ms

Total blocks =1000

(i) Total Encryption Time
DES

$$1000 \times 2 = 2000ms$$

=2 seconds

3-DES

$$1000 \times 6 = 6000ms$$

=6 seconds

Blowfish

$$1000 \times 1 = 1000ms$$

=1 second

Final Table

Algorithm	**Total Time**
DES	2 s
3-DES	6 s
Blowfish	1 s

(ii) Trade-offs

Algorithm	**Security**	**Speed**
DES	Weak (56-bit key)	Fast
3-DES	Stronger than DES	Slow
Blowfish	Strong (128-bit key)	Fastest

(iii) Recommendation
Best choice: Blowfish
Reasons:

- Strong security
- Fastest encryption
- Lower computation
- Suitable for real-time communication

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4. Diffie–Hellman Key Exchange Analysis
Given

- $p = 23$
- $g = 5$
- A's private key =6
- B's private key =15

(d) Compute Public Keys
User A

$$A = 5^6 23$$
$$5^6 = 15625$$
$$15625 23 = 8$$

Public Key A

$$\boxed{8}$$

User B

$$B = 5^{15}\ 23$$

Using modular arithmetic,

$$5^{15}\ 23 = 19$$

Public Key B

$$\boxed{19}$$

(e) Shared Secret
User A

$$K = 19^{6}\ 23$$
$$= 2$$

User B

$$K = 8^{15}\ 23$$
$$= 2$$

Shared Secret

$$\boxed{2}$$

(f) Man-in-the-Middle (MITM) Attack
An attacker intercepts the exchanged public keys:

1. A sends **8**, attacker intercepts it and sends a fake public key to B.
2. B sends **19**, attacker intercepts it and sends another fake public key to A.
3. A and B each establish a shared key with the attacker instead of with each other.
4. The attacker decrypts, reads, modifies, and re-encrypts messages, while A and B believe they are communicating securely.

(g) Prevention

- Use **Digital Signatures** to authenticate public keys.
- Use **Digital Certificates (PKI)** issued by trusted Certificate Authorities.
- Authenticate the exchanged public keys before computing the shared secret.
- Protocols such as **TLS** combine Diffie–Hellman with authentication to prevent MITM attacks.

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5. ECC vs RSA Comparison
Given

- RSA key =1024 bits
- ECC key =160 bits
- IoT device has limited resources

(d) Computational Cost

RSA	**ECC**
Large key operations	Smaller key operations
Slower encryption/decryption	Faster computations
Higher CPU usage	Lower CPU usage

Result: ECC requires less computation.

(e) Power and Memory Consumption

Feature	**RSA**	**ECC**

Feature	RSA	ECC
Key Size	1024 bits	160 bits
Memory	High	Low
Power	Higher	Lower
CPU Usage	Higher	Lower

ECC consumes significantly less memory and battery power, making it ideal for constrained devices.

(f) Best Cryptosystem for IoT
Recommendation: ECC (Elliptic Curve Cryptography)
Reasons:

- Provides security comparable to 1024-bit RSA with only a 160-bit key.
- Faster key generation and cryptographic operations on resource-constrained hardware.
- Lower memory usage and storage requirements.
- Lower power consumption, extending battery life.
- Well suited for IoT devices where processing capability, memory, and energy are limited.

Conclusion: For IoT and other embedded systems, **ECC is the preferred cryptosystem** because it delivers strong security with much lower computational and resource overhead than RSA.